

Image and Voice Pipelines

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Speech pipelines and multimodal evidence. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Speech pipelines and multimodal evidence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the transcription limit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

6. Solve the given case

A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 26 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

If voice or image input is added, show each transformation and a safe fallback.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the transcription limit.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Image and Voice Pipelines

A fictional librarian dictates a shelf note that passes through transcription and catalogue lookup.

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

Image and Voice Pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Audio input

Known phrase names fourteen, not forty.



2. Transcription

Text output changes the shelf number to forty.



3. Cross-check

The authoritative inventory gives shelf 14.



4. Review and correct

Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

CHANGE ONE THING

Remove access to the audio, while keeping the authoritative inventory.

Explain the changed result using the matching lesson.

Which stage first fails, and why should the faulty transcript not be silently overwritten?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian dictates a shelf note that passes through transcription and catalogue lookup.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

1. Audio input
2. Review and correct
3. Transcription
4. Cross-check

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Review and correct Replay the audio, preserve the faulty transcript and use confirmed shelf 14. A correct later lookup cannot repair an earlier wrong input by magic. Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Practice 2: explain the deciding evidence

Changed-case practice: Remove access to the audio, while keeping the authoritative inventory. Which conclusion is supported by the supplied facts?

1. Shelf 14 remains supported for the lookup, but transcription accuracy cannot be replay-checked.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Shelf 14 remains supported for the lookup, but transcription accuracy cannot be replay-checked. The inventory resolves the shelf task; it does not recreate unavailable audio.

Repair a missing building block

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
